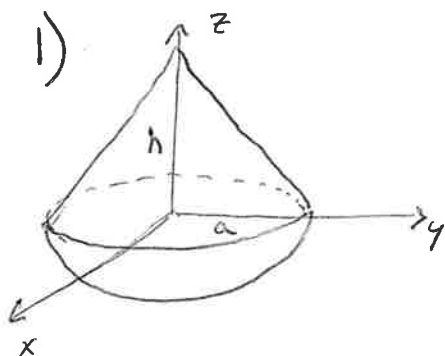


PHY 615 PROBLEM SET 1 (SOLUTIONS)

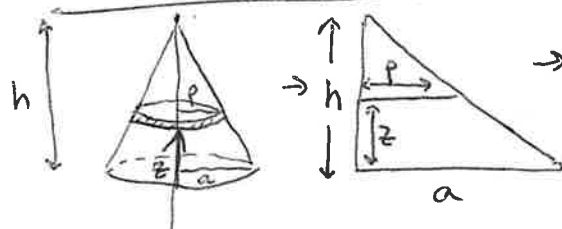
1)



→ determine CM for cone and hemisphere, separately

→ combine the two results using $\vec{R} = \frac{1}{M} \sum \vec{r}_i m_i$

C.M of cone



→ z is dependent on p $\frac{h}{a} = \frac{(h-z)}{p}$ → slopes of similar triangles are same

$$\rightarrow p = (h-z) \cdot \frac{a}{h}$$

$$R_{\text{cone}} = \frac{1}{M} \int z dm, \quad dm = \rho_m dV, \quad \rho_m = \frac{M}{V} = \frac{M}{\frac{1}{3}(\pi a^2)h}$$

$$dV = \pi p^2 dz$$

$$\Rightarrow R_{\text{cone}} = \frac{\rho_m}{M} \int z (\pi p^2) dz$$

$$= \frac{\rho_m \pi}{M} \int_0^h z (h^2 + z^2 - 2hz) \cdot \frac{a^2}{h^2} dz$$

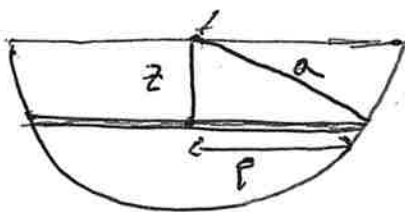
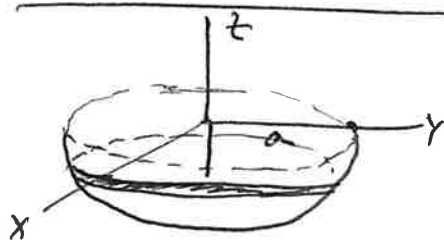
$$= \frac{3M}{\pi a^2 h} \frac{\pi}{M} \frac{a^2}{h^2} \left[\frac{z^2 h^2}{2} + \frac{z^4}{4} - \frac{2hz^3}{3} \right]_0^h$$

$$= \frac{3}{h^3} \left[\frac{h^4}{2} + \frac{h^4}{4} - \frac{2}{3} h^4 \right] = \frac{3}{h^3} \left[\frac{6h^4 + 3h^4 - 4h^4}{12} \right] h^4$$

$$= \frac{3}{h^3} \cdot \frac{h^4}{12} \rightarrow \boxed{R_{\text{cone}} = \frac{h}{4}}$$

1) continued.

CM. of Hemisphere



$$a^2 = z^2 + r^2$$

$$\Rightarrow r^2 = a^2 - z^2$$

$$R_{hs} = \frac{1}{M} \int z dm, \quad dm = \rho_m dV, \quad \rho_m = \frac{M}{V} = \frac{M}{\frac{1}{2} \left(\frac{4}{3} \pi a^3 \right)}$$

$$dV = \pi r^2 dz$$

$$= \frac{1}{M} \int_0^a z \rho_m dV = \frac{M}{V} \frac{1}{M} \int_0^a z \pi r^2 dz$$

$$= \frac{\pi}{V} \int_0^a z (a^2 - z^2) dz = \frac{3\pi}{2\pi a^3} \left[\frac{z^2 a^2}{2} - \frac{z^4}{4} \right]_0^a$$

$$= \frac{3}{2a^3} \left[\frac{2 \cdot a^4}{2} - \frac{a^4}{4} \right] = \frac{3}{2a^3} \frac{a^4}{4}$$

$$\Rightarrow R_{hs} = -\frac{3}{8} a$$

$$\vec{R}_{cm_{tot}} = \frac{1}{(M_{cone} + M_{hs})} (M_{cone} \vec{R}_{cone} + M_{hs} \vec{R}_{hs})$$

$$= \frac{1}{M_{tot}} \left[\frac{M_{cone} h}{4} - \frac{M_{hs} 3a}{8} \right]$$

Now, $\rho_{cone} \times \frac{1}{3} \pi a^2 h = M_{cone}$

$\rho_{hs} \times \frac{2}{3} \pi a^3 = M_{hs}$

$$M_{tot} = \frac{1}{3} \pi a^2 h \rho_c + \frac{2}{3} \pi a^3 \rho_{hs}$$

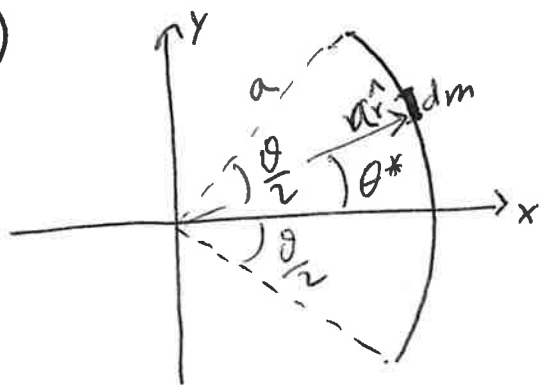
$$= \frac{\pi a^2}{3} [h \rho_c + 2a \rho_{hs}]$$

1) Continued...

$$R_{\text{tot}} = \frac{1}{M_{\text{tot}}} \left[\frac{\rho_c \pi a^2 h^2}{12} - \frac{\rho_{\text{ns}} 3\pi a^4}{12} \right]$$
$$= \frac{3}{\pi a^2 [h \rho_c + 2a \rho_{\text{ns}}]} \cdot \left[\frac{\rho_c \pi a^2 h^2 - \rho_{\text{ns}} 3\pi a^4}{12} \right]$$

$$\Rightarrow \boxed{R_{\text{tot}} = \frac{h^2 \rho_c - 3a^2 \rho_{\text{ns}}}{4(h \rho_c + 2a \rho_{\text{ns}})} \hat{z}}$$

2)



$$\vec{R}_{cm} = \frac{1}{M} \int \vec{r} dm$$

$$\lambda \equiv \frac{dm}{a d\theta} = \frac{M}{a\theta} \rightarrow dm = \lambda a d\theta$$

$$\Rightarrow \vec{r} = a \hat{r} = a (\cos \theta^* \hat{x} + \sin \theta^* \hat{y})$$

$$\vec{R}_{cm} = \frac{\lambda a}{M} \int \vec{r} d\theta = \frac{\lambda a^2}{M} \int (\cos \theta^* \hat{x} + \sin \theta^* \hat{y}) d\theta^*$$

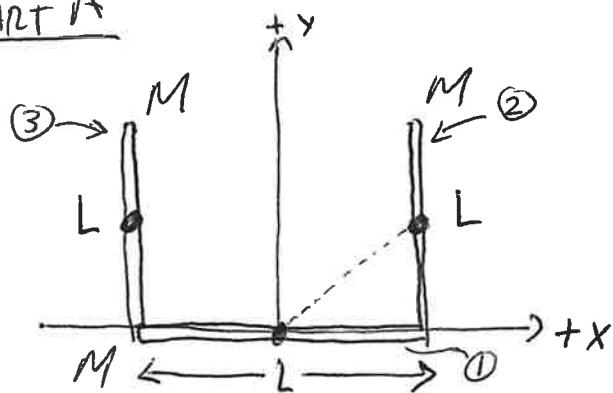
$$\begin{aligned} \int_{-\theta/2}^{\theta/2} \cos \theta^* d\theta^* &= \left. \sin \theta^* \right|_{-\theta/2}^{\theta/2} = \sin \frac{\theta}{2} - \sin \left(-\frac{\theta}{2}\right) = 2 \sin \frac{\theta}{2} \\ \int_{-\theta/2}^{\theta/2} \sin \theta^* d\theta^* &= \left. -\cos \theta^* \right|_{-\theta/2}^{\theta/2} = -\cos \frac{\theta}{2} + \cos \left(-\frac{\theta}{2}\right) = 0 \end{aligned}$$

$$\vec{R}_{cm} = \frac{\lambda a^2}{M} \cdot 2 \sin \frac{\theta}{2} \hat{x}$$

$$= \frac{2M}{a\theta} \cdot \frac{a^2}{M} \sin \frac{\theta}{2} \hat{x} \Rightarrow$$

$$\boxed{\vec{R}_{cm} = \frac{2a}{\theta} \sin \frac{\theta}{2} \hat{x}}$$

3) PART A



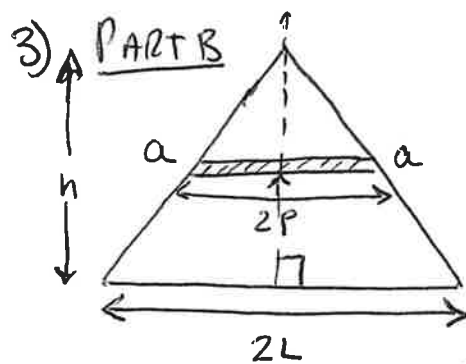
By symmetry, R_{cm} of each rod is at its geometrical center.

$$\vec{R}_{cm1} = 0, \quad \vec{R}_{cm2} = \frac{L}{2} \hat{x} + \frac{L}{2} \hat{y}, \quad \vec{R}_{cm3} = -\frac{L}{2} \hat{x} + \frac{L}{2} \hat{y}$$

$$\vec{R}_{cm} = \frac{1}{3M} \left[M \vec{R}_{cm1} + M \vec{R}_{cm2} + M \vec{R}_{cm3} \right]$$

$$= \frac{1}{3} \left[\cancel{\frac{L}{2} \hat{x}} + \frac{L}{2} \hat{y} - \cancel{\frac{L}{2} \hat{x}} + \frac{L}{2} \hat{y} \right]$$

$$\boxed{\vec{R}_{cm} = \frac{1}{3} L \hat{y}}$$



Find C.M. of an isosceles right triangle

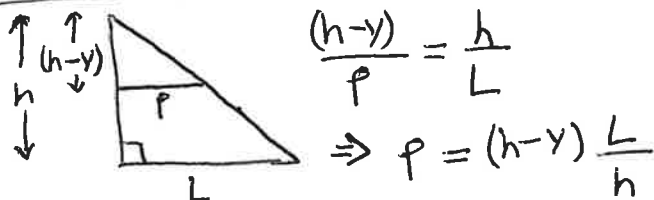
By symmetry the C.M. is in the vertical axis passing through the vertex at half base

Select a slice of mass dm at a height y

$$\vec{R} = \frac{1}{M} \int \vec{r} dm$$

$$\sigma_m \equiv \frac{dm}{dA} = \frac{M}{\frac{1}{2}(2L \cdot h)} = \frac{M}{Lh}$$

$$R_y = \frac{\sigma_m}{M} \int y dA, \quad dA = (2p) dy \rightarrow \text{determine } p \text{ by use of similar triangles}$$



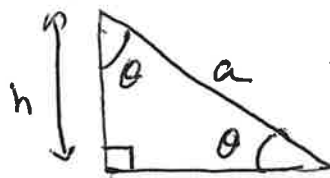
$$= \frac{\sigma_m}{M} \frac{2L}{h} \int_0^h y(h-y) dy$$

$$\frac{y^2 h}{2} - \frac{y^3}{3} \Big|_0^h = \frac{h^3}{2} - \frac{h^3}{3} = \frac{3h^3 - 2h^3}{6} = \frac{h^3}{6}$$

$$= \frac{M}{Lh} \frac{2L}{Mh} \frac{h^3}{6} = \frac{h}{3}$$

$$\Rightarrow \vec{R}_{y_{cm}} = \frac{h}{3} \hat{y}$$

Now, since this is a right isosceles triangle, we have:



$$\Rightarrow h = a \sin \theta$$

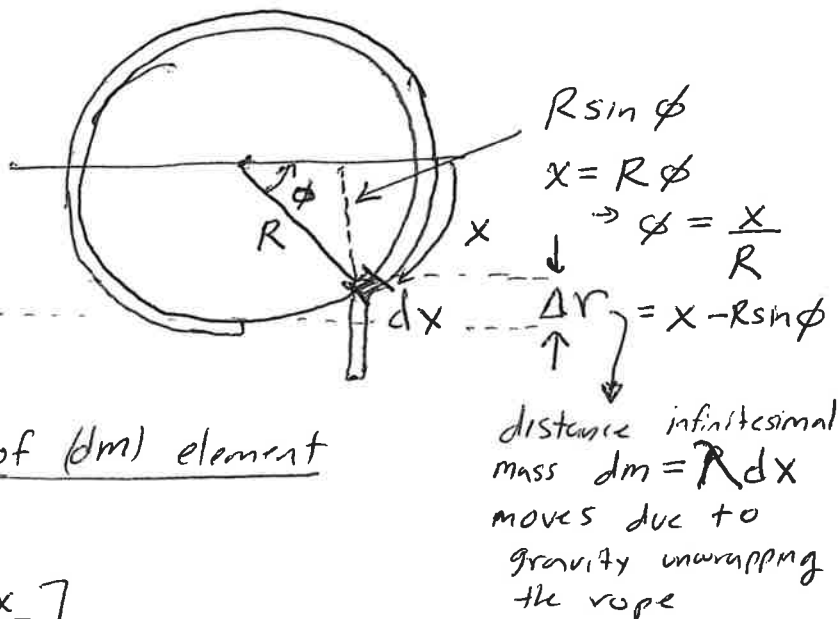
$$\theta + \theta + \frac{\pi}{2} = \pi$$

$$2\theta = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{4}$$

$$h = a \sin \frac{\pi}{4} = \frac{a}{\sqrt{2}}$$

$$\Rightarrow \boxed{\vec{R}_{cm,y} = \frac{a}{3\sqrt{2}} \hat{y}}$$

A diagram of a circular lamina of mass M and radius R , pivoted at point P on its circumference. A horizontal line passes through P . A small element of mass $dm = \lambda dx$ is shown at a distance x from this horizontal line. The angle θ is indicated between the vertical line through P and the line connecting P to the element.


$$dW = F_g \Delta r = (dm) g \Delta r$$

$$\int dW = \int_{RA} \lambda dx \, 2 \left[x - R \sin \frac{x}{R} \right]$$

$$W = \lambda g \int_{x=0}^{R\theta} \left[x - R \sin \frac{x}{R} \right] dx = \lambda g \left[\frac{x^2}{2} + R^2 \cos \frac{x}{R} \right]_{x=0}^{x=R\theta}$$

$$= \lambda g \left[\frac{R^2 \theta^2}{2} + R^2 \cos \theta - R^2 \right] \Rightarrow \boxed{W = \lambda g R^2 \left[\frac{\theta^2}{2} + \cos \theta - 1 \right]}$$

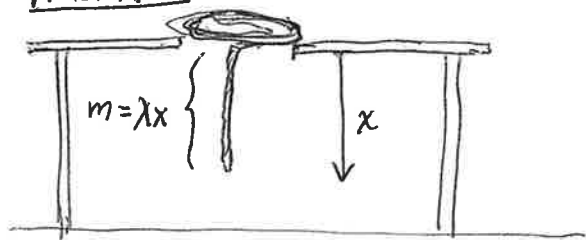
By work energy theorem $\rightarrow \Delta W = \Delta T$ T : Kinetic energy

$$T_{\text{cylinder}} = \frac{1}{2} M (R \dot{\theta})^2 \quad T_{\text{rope}} = \frac{1}{2} m (R \dot{\theta})^2 \quad \lambda = \frac{m}{2\pi R}$$

$$\Delta W = \Delta T \Rightarrow \frac{m g R^2}{2 \pi R} \left[\frac{\theta^2}{2} + \cos \theta - 1 \right] = \frac{1}{2} (M + m) R^2 \dot{\theta}^2$$

$$\Rightarrow \theta^2 = \frac{m g}{\pi R (M+m)} \left[\frac{\theta^2}{2} + \cos \theta - 1 \right]$$

5) PART A



As the rope falls, both its mass m and distance x change, so we must consider a change in momentum with time $\Rightarrow F_{\text{net}} = \frac{dp}{dt}$

At some time $t > 0$, $p = mv = \lambda x v$

The gravity force acting on a section x of the rope is: $F_{\text{net}} = mg = \lambda x g$

$$\Rightarrow \lambda x g = \frac{d}{dt}(mv) = \frac{dm}{dt}v + m \frac{dv}{dt}$$

$$\lambda x g = \lambda \frac{dx}{dt} v + m \frac{dv}{dt} = \lambda v^2 + \lambda x \frac{dv}{dt}$$

$$\rightarrow \text{Consider } v(x(t)) \rightarrow \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx}$$

$$\Rightarrow \lambda x g = \lambda v^2 + \lambda x v \frac{dv}{dx}$$

$$\Rightarrow \boxed{xg = v^2 + x v \frac{dv}{dx}}$$

PART B

Try solution: $v(x) = a x^n \rightarrow \frac{dv}{dx} = a n x^{n-1}$

$$\Rightarrow xg = a^2 x^{2n} + x a x^n \cdot a n x^{n-1}$$

$$= a^2 x^{2n} + a^2 n \cdot \underbrace{x^{(n+1)+(n-1)}}_{x^{2n}}$$

$$\boxed{xg = a^2 x^{2n} (n+1)}$$

$\Rightarrow$ To satisfy this eq. the exponent and coefficients of x must be the same on both sides

$$\Rightarrow 2n = 1 \rightarrow n = \frac{1}{2}$$

$$g = a^2 (n+1) \rightarrow g = \frac{3}{2} a^2 \rightarrow a = \sqrt{\frac{2g}{3}}$$

5) PART B (continued...)

Substituting $n = \frac{1}{2}$ and $a = \sqrt{\frac{2g}{3}}$ into the trial solution

$$\Rightarrow v(x) = ax^n = \sqrt{\frac{2g}{3}} x^{\frac{1}{2}} \Rightarrow \boxed{v(x) = \sqrt{\frac{2gx}{3}}}$$

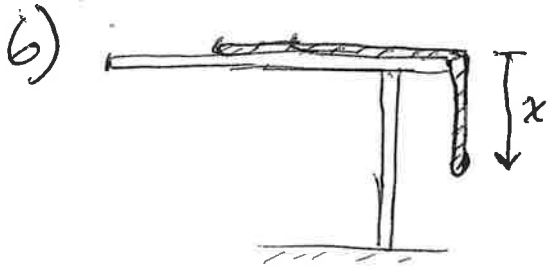
To determine acceleration, $\frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt}$

$$= \sqrt{\frac{2g}{3}} \cdot \frac{1}{2} x^{-\frac{1}{2}} \cdot v$$

$$= \frac{1}{2} \sqrt{\frac{2g}{3x}} \sqrt{\frac{2gx}{3}} = \frac{1}{2} \sqrt{\frac{4g^2}{9}}$$

$$= \frac{1}{2} \frac{2g}{3}$$

$$\Rightarrow \boxed{a = \frac{1}{3}g}$$



$$\lambda = \frac{dm}{dx} = \frac{m}{L}$$

initial conditions

$$t=0, x(t=0) = x_0$$

$$v_x(t=0) = 0$$

* the entire rope of mass m will accelerate

PART A

$$F_{\text{net}} = m \frac{d^2 x}{dt^2} \Rightarrow \left(\frac{m}{L}\right) x \cdot g = m \frac{d^2 x}{dt^2} \Rightarrow \boxed{\frac{d^2 x}{dt^2} = \frac{gx}{L}}$$

$$\text{let } \omega^2 = \frac{g}{L}$$

Try a linear combination $x(t) = A e^{\omega t} + B e^{-\omega t}$

$$\begin{aligned} \rightarrow \frac{dx}{dt} &= \omega A e^{\omega t} - \omega B e^{-\omega t} & \frac{d^2 x}{dt^2} &= \omega^2 A e^{\omega t} + \omega^2 B e^{-\omega t} \\ & & &= \omega^2 (A e^{\omega t} + B e^{-\omega t}) \quad \checkmark \end{aligned}$$

to determine the const. A & B :

$$x(t=0) = A + B = x_0$$

$$v = \frac{dx}{dt}(t=0) = \omega(A - B) = 0 \Rightarrow \underline{A=B} \Rightarrow A+B=2A=x_0$$

$$\underline{\underline{A = \frac{x_0}{2}, B = \frac{x_0}{2}}}$$

$$\Rightarrow \boxed{x(t) = \frac{x_0}{2} (e^{\omega t} + e^{-\omega t})}$$

Alternatively, using the relation: $\cosh u = \frac{1}{2}(e^u + e^{-u})$

$$\Rightarrow \boxed{x(t) = x_0 \cosh \omega t}$$

PART B

$$\text{Solving for } t: \cosh^{-1}\left(\frac{x(t)}{x_0}\right) = \omega t$$

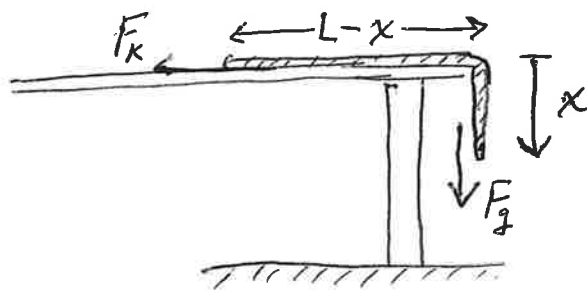
$$\rightarrow t(x) = \frac{1}{\omega} \cosh^{-1}\left(\frac{x(t)}{x_0}\right),$$

$$\Rightarrow \boxed{t^* = \frac{1}{\omega} \cosh^{-1}\left(\frac{L}{x_0}\right)}$$

when (left) end of rope reaches edge of table
 $x(t^*) = L$

where t^* is the time to reach edge

7)



$$\lambda = \frac{m}{L}$$

top part of rope



$$N = \lambda(L-x)g$$

PART A

$$i) F_g = (\lambda x)g$$

$$ii) F_k = \mu_k N = \mu_k g \lambda(L-x)$$

$$iii) F_{net} = F_g - F_k = \lambda xg - \mu_k g \lambda(L-x)$$

$$F_{net} = \lambda g [x - \mu_k(L-x)]$$

* Since rope is given a pull, we assume that $F_g > F_k$ for rope to accelerate down

PART B

To get the rope to start moving, $F_g \geq F_k$

The minimum value $x = x_0 \Rightarrow F_g = F_k$

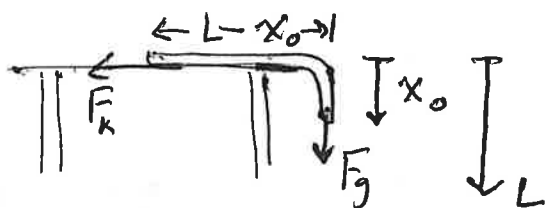
$$\Rightarrow \cancel{\lambda} x_0 g = \mu_k \cancel{\lambda} g (L - x_0)$$

$$x_0 = \mu_k L - \mu_k x_0 \Rightarrow x_0(\mu_k + 1) = \mu_k L$$

$$\Rightarrow \boxed{x_0 = \frac{\mu_k L}{\mu_k + 1}}$$

Minimum length of rope hanging required to start moving

7) PART C



By the work-energy theorem, the net work done on the rope equals a change in the kinetic energy.

$$\Rightarrow W_{\text{net}} = \Delta K, \quad W_{\text{net}} = \int F_{\text{net}} \cdot dx$$

$$W_{\text{net}} = \int \lambda g [x - M_k (L - x)] dx$$

$$= \lambda g \left[\frac{x^2}{2} - M_k (Lx - \frac{x^2}{2}) \right]_{x_0}^L$$

$$= \lambda g \left[\frac{L^2}{2} - M_k \left(L^2 - \frac{L^2}{2} \right) - \frac{x_0^2}{2} + M_k (Lx_0 - \frac{x_0^2}{2}) \right]$$

$$= \lambda g \left[\frac{L^2}{2} - \frac{1}{2} M_k L^2 - \frac{x_0^2}{2} + M_k Lx_0 - \frac{1}{2} M_k x_0^2 \right]$$

$$= \lambda g \left[\frac{L^2}{2} (1 - M_k) - \frac{x_0^2}{2} (M_k + 1) + M_k Lx_0 \right], \quad \text{Recall:}$$

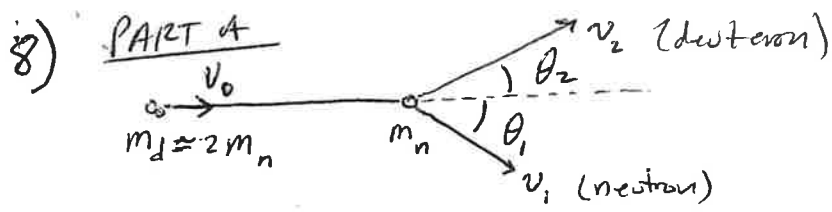
$$= \lambda g \left[\frac{L^2}{2} (1 - M_k) - \frac{M_k^2 L^2 (M_k + 1)}{2(M_k + 1)^2} + \frac{2 \cdot M_k L^2}{2 \cdot (M_k + 1)} \right]$$

$$= \lambda g \left[\frac{L^2 (1 - M_k)(M_k + 1)}{2(M_k + 1)} + \frac{M_k^2 L^2}{2(M_k + 1)} \right] = \frac{\lambda g}{2(M_k + 1)} \left[-L^2 (M_k^2 - 1) + M_k^2 L^2 \right]$$

$$\Rightarrow W_{\text{net}} = \frac{\lambda g L^2}{2(M_k + 1)} = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \xrightarrow{v_i = 0} -M_k^2 L^2 + L^2 + M_k^2 L^2$$

$$\Rightarrow \frac{1}{2} m v_f^2 = \frac{m}{2} \frac{g L^2}{(M_k + 1)}$$

$$\xrightarrow{\Delta K} \boxed{v_f = \sqrt{\frac{g L}{M_k + 1}}}$$



Given

$$v_0 = 14.9 \text{ km/s}$$

$$\theta_2 = 10^\circ$$

Energy Conservation

$$\frac{1}{2} m_d v_0^2 = \frac{1}{2} m_d v_2^2 + \frac{1}{2} m_n v_1^2 \rightarrow \frac{1}{2} (2m_n) v_0^2 = \frac{1}{2} (2m_n) v_2^2 + \frac{1}{2} m_n v_1^2$$

$$\Rightarrow m_n v_0^2 = m_n v_2^2 + \frac{1}{2} m_n v_1^2$$

$$v_0^2 = \frac{1}{2} v_1^2 + v_2^2$$

Momentum Conservation

$$\left. \begin{aligned} 2m_n v_0 &= 2m_n v_2 \cos \theta_2 + m_n v_1 \cos \theta_1 \\ 2m_n v_2 \sin \theta_2 &= m_n v_1 \sin \theta_1 \end{aligned} \right\} \begin{array}{l} \text{Solve to} \\ \text{eliminate } \theta_1 \text{ and} \\ \text{obtain eq. only in terms} \\ \text{of } v_1, v_2 \text{ \& } \cos \theta_2 \end{array}$$

Let $m_n \equiv m$ (no subscript, for simplicity)

$$\Rightarrow m(v_1 \cos \theta_1)^2 = (2m v_0 - 2m v_2 \cos \theta_2)^2$$

$$+ m(v_1 \sin \theta_1)^2 = (2m v_2 \sin \theta_2)^2$$

$$v_1^2 (\underbrace{\sin^2 \theta_1 + \cos^2 \theta_1}_{=1}) = 4v_0^2 + 4v_2^2 \cos^2 \theta_2 - 8v_0 v_2 \cos \theta_2 + 4v_2^2 \sin^2 \theta_2$$

$$v_1^2 = 4v_0^2 - 8v_0 v_2 \cos \theta_2 + 4v_2^2 (\underbrace{\sin^2 \theta_2 + \cos^2 \theta_2}_{=1})$$

Combining Energy / Momentum conservation:

$$\Rightarrow \left\{ \begin{aligned} v_1^2 &= 2(v_0^2 - v_2^2) \\ v_1^2 &= 4v_0^2 + 4v_2^2 - 8v_0 v_2 \cos \theta_2 \end{aligned} \right. \begin{array}{l} 2 \text{ Equations} \\ 2 \text{ Unknowns } (v_1, v_2) \end{array}$$

↳ These eqs. can be solved for v_1 & v_2

$$\boxed{v_1 = 5.18 \text{ km/s}} \quad \text{or} \quad \boxed{v_1 = 19.79 \text{ km/s}}$$

$$\boxed{v_2 = 14.44 \text{ km/s}} \quad \text{or} \quad \boxed{v_2 = 5.12 \text{ km/s}}$$

8) continued...

PART B : To solve for θ_1 (newton scatt. angle)

Use : $2n_1 v_2 \sin \theta_2 = n_1 v_1 \sin \theta_1 \rightarrow (v_1 \& v_2 \text{ obtained from part 1, } \theta_2 \text{ is given})$

Solve for θ_1 :

$$\Rightarrow \sin \theta_1 = \frac{2v_2 \sin \theta_2}{v_1}$$

$$\Rightarrow \theta_1 = \sin^{-1} \left[\frac{2v_2}{v_1} \sin \theta_2 \right]$$

$$\theta_1 = 74.84^\circ \quad \text{or} \quad \theta_1 = 5.16^\circ$$